Transductive Prototypical Attention Reasoning Network for Few-Shot SAR Target Recognition

Haohao Ren, *Member, IEEE*, Sen Liu, *Student Member, IEEE*, Xuelian Yu, *Member, IEEE*, Lin Zou, *Member, IEEE*, Yun Zhou, Xuegang Wang, and Hao Tang

***Abstract*— Deep-learning-based synthetic aperture radar (SAR) automatic target recognition (ATR) algorithms have achieved outstanding performance under the condition of hundreds or thousands of training samples in recent years. Nevertheless, it is often rare to acquire great quantities of target samples in real SAR application scenarios. This article proposes a novel ATR method called transductive prototypical attention reasoning network (TPARN) to solve the problem of SAR target recognition with only a few training samples. To be specific, a region-awareness-based feature extraction model is first developed, which can effectively focus on the target region of interest (RoI) and suppress the background clutter by embedding direction-aware and position-sensitive information to extract more transferable knowledge. To heighten the discrimination of the sample features, a cross-feature spatial attention (CFSA) module is then proposed following the feature embedding model. Finally, a transductive prototype reasoning method is presented to realize the identity reasoning of the target, which can continuously update each class prototype with training samples and test samples together, thereby improving the classification accuracy. In addition, a marginal adaptive hybrid loss (MAHL) is proposed to obtain a discriminative feature embedding space with intraclass compactness and interclass divergence, aiming to facilitate subsequent target identity reasoning. Extensive experiments on the moving and stationary target acquisition and recognition (MSTAR) benchmark dataset reveal that the proposed method outperforms some state-of-the-arts under different few-shot SAR ATR tasks.**

***Index Terms*— Automatic target recognition (ATR), convolutional neural network (CNN), few-shot learning (FSL), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor with high-resolution, day-and-night, and all-weather operating characteristics [1], [2], which has played a decisive role in various fields, i.e., military defense, environmental monitoring, resource exploration, mapping, and so on. As one of the important SAR image interpretation techniques, automatic target recognition (ATR) has been attracting much attention. After years of exploration and efforts, SAR ATR techniques have made remarkable advancement. In early days, SAR ATR methods rely on hand-crafted features such as target shadow [3], [4], nonnegative features [5], monogenic features [6], [7], sparse features [8], [9], [10], [11], and scattering feature [12], [13], which are derived from the empirical knowledge and reasoning of the experts, resulting in poor generalization and robustness.

In the past few years, deep learning has received extensive attention in various fields because it can autonomously extract deep semantic information from large volume of data in an end-to-end way. In the field of SAR image interpretation, a variety of deep-learning-based ATR methods have been proposed in succession recently. Chen et al. [14] applied deep convolutional neural network (CNN) to SAR target recognition task for the first time, which laid a foundation for the later research on deep-learning-based SAR target recognition methods. Zhang et al. [15] proposed a new feature fusion framework for SAR target recognition, which makes full use of the advantages of electromagnetic scattering features and deep CNN features to improve the target recognition performance. Tao et al. [16] presented a novel CNN-based dictionary learning method to achieve robust SAR target recognition. Kechagias-Stamatis and Aouf [17] proposed a multimodal fusion strategy integrating the merits of CNN and sparse representation, which can achieve excellent SAR target recognition performance under multiple operation conditions.

As we all know, there is a frustrating situation that these deep-learning-based SAR target recognition methods can only achieve excellent recognition performance in the presence of large quantities of available labeled samples. Nevertheless, acquiring quite a few of SAR images in real application scenarios is expensive or even impossible. To alleviate the dependence of the recognition model on a great many training samples, scholars have recently put considerable effort into achieving robust recognition of SAR targets with limited training samples. To name just a few, Pei et al. [18] proposed a multiview generation method to augment training samples and developed a parallel multiview CNN for SAR target

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Haohao Ren, Sen Liu, Xuelian Yu, Lin Zou, Yun Zhou, and Xuegang Wang are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu 611731, China (e-mail: xlyu2012@uestc.edu.cn).

Hao Tang is with the Department of Information Technology and Electrical Engineering, ETH Zürich, 8092 Zürich, Switzerland (e-mail: hao.tang@vision.ee.ethz.ch).

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recognition with limited training samples. Ding et al. [19] investigated the efficacy of three data augmentation strategies in mitigating the overfitting phenomenon of the CNN-based SAR target recognition models. Lin et al. [20] presented a deep neural network model embedded with convolutional highway layers to solve SAR target recognition under the condition of limited training samples. Sun et al. [21] designed an angular rotation generative network to achieve SAR target recognition in the case of limited training samples. Although these ATR methods mentioned above can alleviate the dependence of recognition model on a great many training samples to some extent, it still requires at least dozens of samples for each category of target to achieve robust recognition performance.

Inspired by human cognitive mechanism, few-shot learning (FSL) has been a hot topic in the field of machine learning for the past few years. Different from other learning paradigms, FSL is dedicated to leveraging learned prior knowledge to quickly generalize to new tasks with a few training samples [22]. It is worth noting that the idea of transfer learning and meta-learning is often applied in FSL tasks. Because the idea of meta-learning is closer to the human cognitive mechanism, it has become a research hotspot to solve FSL problems at this stage. Up to now, a variety of FSL methods integrating the idea of meta-learning have been proposed successively. Among them, prototypical network (ProtoNet) is a classic FSL method in the field of computer vision [23]. Relation Network (RelationNet) is an end-to-end FSL method, which reasons the identity of the unknown target by learning deep similarity instead of fixed metric [24].

Moreover, graph neural network (GNN)-based FSL methods such as transductive propagation network (TPN) [25], [26] and edge-labeling GNN [27] have also been proposed in the past few years. Unlike other FSL methods, GNN-based methods usually leverage the idea of transductive learning to reason the identity of the target. To boost the learning ability of the FSL model, scholars have invested considerable effort in the loss function of the model. For instance, two contrastive losses including triplet loss and center loss are exploited to improve the discrimination of embedding space for FSL tasks [28]. Ren et al. [29] designed a bimargin loss to promote the intrarelation compactness and interrelation dispersion for the feature representation in a semantic space for few-shot relation classification. Shao et al. [30] proposed a supervised contrastive loss to improve the generalization of model-agnostic meta-learning (MAML).

With regard to the problem of SAR target recognition with only a few training samples, FSL paradigm provides a feasible and effective scheme. As a consequence, scholars have recently pursued preliminary work in the field of few-shot SAR target recognition [31], [32], [33], [34]. For instance, Wang et al. [35] proposed a ProtoNet-based few-shot SAR ATR method, which exploits the LTSM network instead of the CNN model as feature embedding network. Fu et al. [36] presented a meta-transfer learning framework to realize few-shot SAR target recognition (MSAR). Wang et al. [37] designed a hybrid inference network, which leverages a decision fusion strategy to promote few-shot SAR target recognition performance. Wang et al. [38] proposed a multitask representation learning framework (MTRLF) to achieve few-shot SAR target recognition. Ren et al. [39] presented a Siamese subspace classification network (2SCNet) to solve the problem of few-shot SAR target recognition. Currently, the research on few-shot SAR target recognition is still in its infancy, and there exist three shortcomings in these few-shot SAR target recognition methods. First, most of the existing few-shot SAR target recognition methods pay too much attention to the reasoning stage, while ignoring the importance of the feature extraction model, resulting in unsatisfactory recognition performance. Second, current few-shot SAR target recognition methods generally follow the paradigm of meta-learning with episodic training to learn a feature extraction model, in which the categories of the training stage and the reasoning stage are disjoint. As pointed out in previous work [40], [41], [42], current methods independently extract features from labeled few-shot training samples and samples to be recognized in the few-show target reasoning stage, so the extracted features are not discriminative enough. Finally, the existing few-shot SAR target recognition methods are all based on inductive learning; as a result, the performance and robustness of these methods are severely dependent on the quality of the class prototype.

To achieve robust few-shot SAR target recognition, this article puts forward a novel few-shot method called transductive prototypical attention reasoning network (TPARN). The main contributions of this article can be summarized in the following fourfolds.

1) A novel method named TPARN is proposed to address the problem of few-show SAR target recognition, which resorts to two aspects, i.e., two-stage attention-based feature extraction model and transductive-based reasoning strategy to promote target recognition performance. Extensive quantitative and qualitative experiments demonstrate that the proposed method is superior to some state-of-the-art few-shot SAR target recognition methods.
2) To more accurately locate the region of interest (RoI) while suppressing the clutter, a region-awareness-based feature extractor is designed, which not only captures cross-channel information but also acquires direction-aware and position-sensitive information. Meanwhile, in terms of the problem that the target categories of the training stage and the reasoning stage are disjoint, which results in weak discrimination of the sample features learned in the reasoning stage, a cross-feature spatial attention (CFSA) module consisting of cross-feature fusion (CFF) and spatial attention generation (SAG) is developed following the feature extraction model to learn feature relations so as to heighten the discrimination of extracted features in the few-shot reasoning stage.
3) To achieve robust few-shot target identity reasoning, a transductive prototype reasoning method is presented, which can promote the quality of class prototypes through the idea of transductive learning, thereby improving the accuracy of target identity reasoning under the condition of limited samples.

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4) A marginal adaptive hybrid loss (MAHL) is proposed, which can force samples to be closer to the same class prototype rather than to other class prototypes by the minimal category margin threshold, aiming to promote intraclass compactness and interclass divergence in the embedding space.

The remainder of this article is organized as follows. Section II introduces related work on meta-learning and attention mechanisms. Section III elaborates each key component of the proposed TPARN. Extensive experiments on the MSTAR benchmark dataset are conducted to illustrate the effectiveness of the proposed method in Section IV. Section V concludes this article.

II. Related Work

A. Meta-Learning-Based FSL

Generally speaking, meta-learning refers to learning to learn, namely, it can guide the model to acquire the ability of "learning to learn," so that the model can quickly learn knowledge in a new task with the help of priori knowledge.

The FSL methods that incorporate the idea of meta-learning usually consist of two stages: meta-training stage and meta-test stage. Datasets for the meta-training and meta-test stages are termed as meta-training and meta-test data, respectively, and it is stipulated that there is no category intersection between the two datasets. The meta-training dataset is divided into support samples and query samples, which are used to enable the model to acquire transferrable knowledge and the ability to "learning to learn." The meta-test dataset contains few training samples and samples to be recognized. To equip the model with sufficient transferrable knowledge, it is usually required that the number of target categories and size of samples in the meta-training set are much greater than those in the meta-test set. Suppose that the total categories of target are N, and only K samples are acquired for each category of target, such FSL task is termed as the N-way K-shot. Different from the training way of traditional models, FSL integrating the idea of meta-learning takes meta-task as the training unit. As a result, a batch of data is composed of multiple meta-tasks instead of multiple samples during model training. Currently, many FSL methods such as ProtoNet [23], RelationNet [24], and TPN [26] incorporate the idea of meta-learning.

B. Attention-Based SAR ATR Methods

The attention mechanism is a burgeoning technique that simulates human cognitive attention in an artificial neural network. Simply put, the function of the attention mechanism is to highlight useful information and suppress useless one in a neural network, thereby improving the learning ability of the network. Of late years, various kinds of attention model have been emerged one after another in the field of computer vision. In particular, these attention models are usually lightweight and plug-and-play. Among them, SENet [43] is one of the most representative attention modules in recent years, which refines extracted features at channel level. Subsequently, other state-of-the-art attention modules such as convolutional block attention module (CBAM) [44], nonlocal neural networks [45], and coordinate attention [46] were proposed to refine features at both the spatial and channel levels, and these methods can extract more discriminative features than SENet. Since SAR target feature extraction is susceptible to the background clutter, the attention mechanism provides a very feasible solution for discriminative features' extraction. Wang et al. [47] applied SENet to SAR target feature extraction for the first time and demonstrated the effectiveness of the attention mechanism for improving SAR ATR performance through extensive experiments. Later, attention modules such as CBAM and cross-interaction attention have been successfully applied to SAR target recognition, and numerous experiments on the MSTAR benchmark dataset demonstrated that they are very beneficial to promote SAR target recognition performance [48], [49], [50], [51].

III. Transductive Prototypical Attention Reasoning Network

In this section, we elaborate the proposed TPARN that is composed of three key components: region-awareness-based feature extractor, CFSA module, and transductive prototype reasoning module, as depicted in Fig. 1. For the few-shot SAR target recognition task, TPARN consists of the meta-training and meta-test phases. In the meta-training phase, a region-awareness-based feature extraction model is first designed to map original high-dimensional SAR images to low-dimensional feature embedding space. Second, a CFSA module is developed to heighten the discrimination of extracted features. Then, a transductive prototype reasoning method is proposed to reason the identity of the SAR target to be recognized. Finally, an MAHL is calculated with the help of the true label to optimize the whole model. In the meta-test phase, the identity of the target is reasoned using the recognition model obtained in the meta-training phase. In what follows, the proposed method is introduced in detail.

A. Region-Awareness-Based Feature Extractor

Feature extraction is one of the essential steps of the SAR target recognition system. Due to its autonomous learning characteristics with an end-to-end manner, CNN has been widely used as the feature embedding model at this stage. However, whether at channel level or at spatial level, CNN treats all the regions in an image equally rather than focusing on the interest of region. This inherent deficiency of CNN results in its limited learning capability.

In practice, target only occupies a few pixels in an SAR image, and target feature extraction is easily disturbed by the background clutter. Drawing lessons from the idea of attention mechanism [43], [44], [45], [46], this article proposes a region-awareness-based feature extractor, which can accurately locate the target RoI while reducing the impact of the background clutter on target feature extraction. An RoI awareness operation based on coordinate attention [46] is developed in the feature extraction model. Different from the classical attention models such as SENet and CBAM, the region-awareness-based feature extractor decomposes the channel attention along the horizontal and vertical directions into two 1-D features' encoding and captures the long-distance

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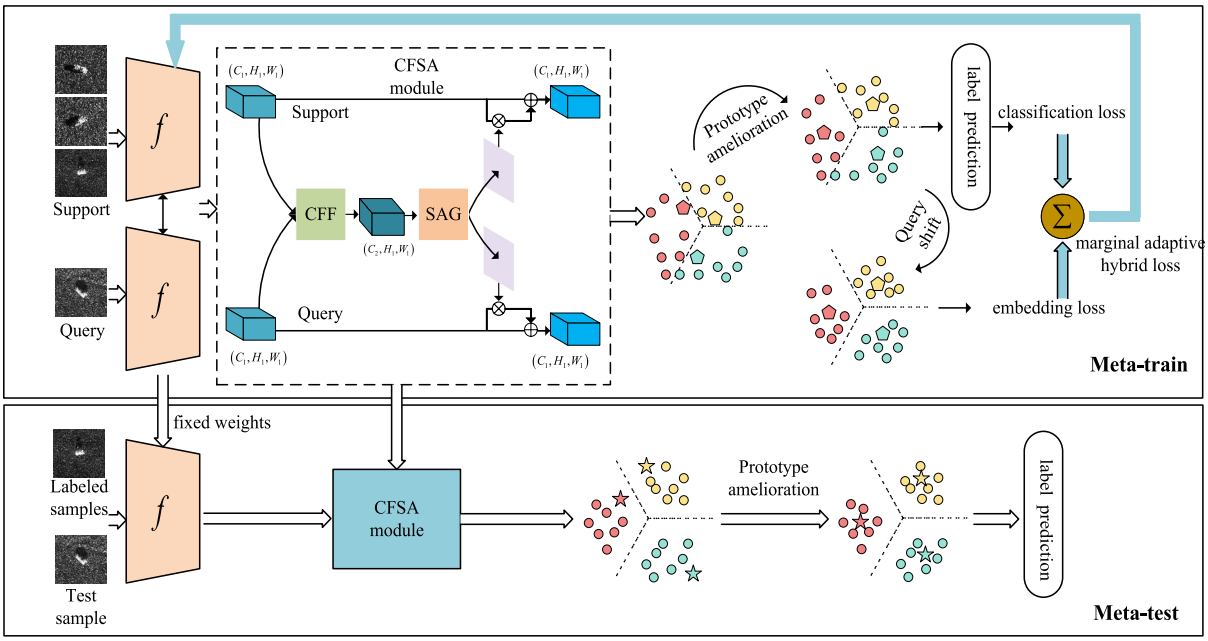


Fig. 1. Framework of the proposed TPARN.

dependencies of the input feature map along different spatial directions, so as to integrate the spatial coordinate information efficiently. By doing so, the region-awareness-based feature extractor not only captures cross-channel information but also learns direction-aware and position-sensitive information, which makes the model more accurately locate the target area while suppressing the background clutter than the existing attention modules. Specifically, let $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ be the convolutional features of an SAR image, where C represents the number of feature channels, and W and H represent the size of the SAR image feature in the horizontal and vertical directions, respectively. We first leverage two average pooling kernels to encode each feature channel along the horizontal and vertical coordinates, respectively. Mathematically, the two average pooling operations can be expressed as follows:

$$\mathbf{F}^{\mathbf{x}} = \frac{1}{W}\sum_{y=1}^{W}\mathbf{F}(x, y) \tag{1}$$

$$\mathbf{F}^{\mathbf{y}} = \frac{1}{H}\sum_{x=1}^{H}\mathbf{F}(x, y) \tag{2}$$

where $\mathbf{F}^{\mathbf{x}} \in \mathbb{R}^{C\times H\times 1}$ represents the 1-D SAR feature in the horizontal direction, and $\mathbf{F}^{\mathbf{y}} \in \mathbb{R}^{C\times 1\times W}$ represents the 1-D SAR feature map in the vertical direction. The above two operations can effectively aggregate SAR target features along the two spatial directions, respectively. By doing so, the proposed method is capable of capturing long-range dependencies along one spatial direction while preserving positional information along another spatial direction, which is very helpful for accurately locating the target RoI in the SAR image.

Next, to obtain an attention map that can be aware of the target RoI in the SAR image, we concatenate the two pooling features obtained in (1) and (2) and feed them into a convolutional layer with the kernel size of 1×1, yielding

$$\mathbf{F_1} = \sigma\left(\mathrm{Conv}\left(\mathrm{Concat}\left(\mathbf{F}^{\mathbf{x}}, \mathbf{F}^{\mathbf{y}}\right)\right)\right) \tag{3}$$

where $\mathbf{F_1} \in \mathbb{R}^{C/r\times(H\times W)}$, Concat$(\cdot)$ denotes the concatenation operation, Conv represents the convolution operation with a convolution kernel of size 7×7, σ is a nonlinear activation function (e.g., ReLU), and r is a compression ratio that is used to reduce the computational cost, as in SENet [43].

Afterward, we split $\mathbf{F}_1$ along two different spatial dimensions to obtain $\mathbf{F}_1^{\mathbf{x}} \in \mathbb{R}^{C/r\times H}$ and $\mathbf{F}_1^{\mathbf{y}} \in \mathbb{R}^{C/r\times W}$. Two convolutions with the kernel size of 1×1 are used to transform $\mathbf{F}_1^{\mathbf{x}}$ and $\mathbf{F}_1^{\mathbf{y}}$ into new features $\mathbf{A}^{\mathbf{x}}$ and $\mathbf{A}^{\mathbf{y}}$ with the same numbers of channels as the original feature $\mathbf{F}$. This step operation is described as follows:

$$\mathbf{A}^{\mathbf{x}} = \delta\left(\mathrm{Conv}\left(\mathbf{F}_1^{\mathbf{x}}\right)\right) \tag{4}$$

$$\mathbf{A}^{\mathbf{y}} = \delta\left(\mathrm{Conv}\left(\mathbf{F}_1^{\mathbf{y}}\right)\right) \tag{5}$$

where δ is the sigmoid activation function. $\mathbf{A}^{\mathbf{x}}$ and $\mathbf{A}^{\mathbf{y}}$ are two attention weights along two directions, respectively.

Finally, the output $\mathbf{F}_{\mathrm{out}}$ can be obtained according to the following operation:

$$\mathbf{F}_{\mathrm{out}} = \mathbf{F} \times \mathbf{A}^{\mathbf{x}} \times \mathbf{A}^{\mathbf{y}}. \tag{6}$$

Compared with the original feature $\mathbf{F}$, $\mathbf{F}_{\mathrm{out}}$ is able to highlight the target RoI while suppressing the background clutter. To accurately locate the target RoI in the SAR image after layer-by-layer convolution operation, the above region awareness operation is embedded behind multiple convolution layers. The architecture of the proposed region-awareness-based feature extractor is displayed in Fig. 2.

B. CFSA Module

It is known that the target categories of the training stage and the reasoning stage in the existing few-shot SAR target

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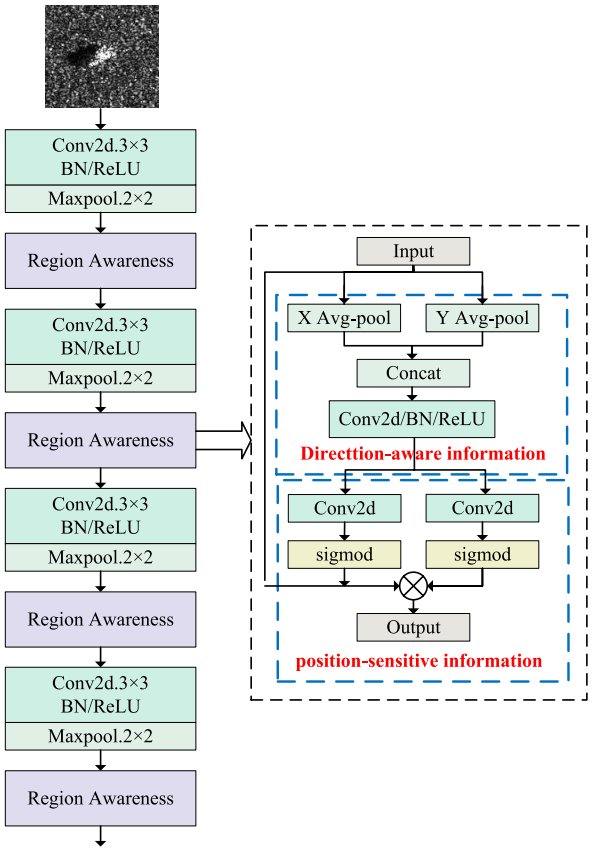


Fig. 2. Network architecture of the region-awareness-based feature extractor.

recognition task are disjoint [23], [31], [34], [38], that is to say, the labeled samples in the reasoning stage do not participate in the training of the feature extraction model. In the reasoning stage, the existing few-shot SAR ATR methods directly use the feature extraction model obtained in the training stage. That is, they extract features from labeled and unlabeled samples independently. Because labeled samples in the reasoning stage are not used for feature extraction model training, the features extracted from labeled and unlabeled samples independently are not discriminative enough. To improve this deficiency, this article proposes a CFSA module to learn feature relations to heighten the discrimination ability of the extracted features. The proposed CFSA module consists of two vital operations: CFF and SAG, as shown in Fig. 1.

Concretely, let $\mathbf{F_S} \in \mathbb{R}^{C_1\times H_1\times W_1}$ and $\mathbf{F_Q} \in \mathbb{R}^{C_1\times H_1\times W_1}$ be the corresponding features of support and query samples, respectively, obtained by feature extraction model, where C_1 is the number of channels of feature map, and W_1 and H_1 represent the length and width of the feature, respectively. First, we develop a fusion strategy to fuse class prototype features obtained from support samples with features of query samples. Drawing on the idea of compact bilinear pooling (CBP) [52], we leverage bilinear pooling [53] to fuse the extracted features, namely,

$$\mathbf{F}_{\text{new}} = B\left(\mathbf{F}_S, \mathbf{F}_Q\right) \tag{7}$$

where $B(\cdot)$ denotes the bilinear pooling operation.

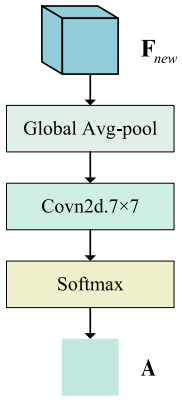


Fig. 3. Network architecture of SAG.

To control the dimension of the fused features, we adopt the kernel machine method in this article. Let $K(\cdot,\cdot)$ be a second-order polynomial kernel. We resort to finding a low-dimensional projection function $\phi(\mathbf{F}_{\text{new}}) \in R^d$, where $d << (C_1H_1W_1)^2$, which satisfies $\langle\phi(\mathbf{F}_S), \phi(\mathbf{F}_Q)\rangle \approx K(\mathbf{F}_S, \mathbf{F}_Q))$, i.e.,

$$\left\langle B\left(\mathbf{F}_S, \mathbf{F}_Q\right), B\left(\mathbf{F}_S, \mathbf{F}_Q\right)\right\rangle \approx \sum\sum\left\langle\phi(\mathbf{F}_S), \phi\left(\mathbf{F}_Q\right)\right\rangle. \tag{8}$$

In this article, Tensor Sketch (TS) function [54] is used to approximate the polynomial kernel. Constrained by the article space, the main processing flow of feature fusion is summarized in Algorithm 1.

Algorithm 1 Algorithm for the CFF Method in the CFSA Module

Input: $\mathbf{F_S} \in \mathbb{R}^{C_1\times H_1\times W_1}$, $\mathbf{F_Q} \in \mathbb{R}^{C_1\times H_1\times W_1}$
Output: $\mathbf{F}_{new} \in \mathbb{R}^{C_2\times H_1\times W_1}$

1: set $D_1 = C_1 \times H_1 \times W_1$, $D_2 = C_2 \times H_1 \times W_1$
2: **for** $a = 1, 2$ **do**
3: Random sampling to get $h_a(b) \in \{1, 2, \cdots, D_2\}$, and $v_a(b) \in \{+1, -1\}^{D_1}$, where $h_a \in \mathbb{N}^{D_1}$, and $v_a(b)$ is uniformly drawn from $\{+1, -1\}$.
4: **end for**
5: Define the sketch function $\psi(\mathbf{F_S}) = \{L_1, L_2, \cdots, L_{D_2}\}$, where $L_j = \sum_{t:h_a(t)=j} v(t)\mathbf{F_s}$. Likewise, get $\psi(\mathbf{F_Q})$
6: Generate $\mathbf{F}_{new} = \text{FFT}^{-1}(\text{FFT}(\psi(\mathbf{F_S})) \odot \text{FFT}(\psi(\mathbf{F_Q})))$, where the $\odot$ denotes elementwise multiplication.

After obtaining the fused feature $\mathbf{F}_{\text{new}}$, we aggregate spatial information of feature $\mathbf{F}_{\text{new}}$ using global-average pooling operation to generate spatial context descriptors: $\mathbf{F}^s_{\text{new}}$. Afterward, $\mathbf{F}^s_{\text{new}}$ is fed into a convolution layer with kernel size of 7×7 to produce initial attention map, yielding

$$\mathbf{F}^a_{\text{new}} = \text{Conv}_{7\times7}(\text{Avg}(\mathbf{F}_{\text{new}})). \tag{9}$$

Eventually, the spatial attention map is generated according to the following formula:

$$\mathbf{A} = \omega\left(\mathbf{F}^a_{\text{new}}\right) \tag{10}$$

where ω represents the softmax function. Fig. 3 depicts the network architecture of the SAG.

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Drawing on the advantages of residual learning, we obtain discriminative features by the following formula:

$$\mathbf{F_S}' = \mathbf{F_S} \times (\mathbf{A} + 1)$$
$$\mathbf{F_Q}' = \mathbf{F_Q} \times (\mathbf{A} + 1). \tag{11}$$

C. Transductive Prototype Reasoning

Classification is the last key component of a target recognition system. Inductive learning and transductive learning are two widely used reasoning methods in a classification model. In simple terms, inductive learning first exploits all the training samples to learn a model that can represent the class distribution, and then use this model to reason the identity of the target to be recognized. Whereas transductive learning is first put together with all the training samples and the samples to be recognized, and then reason the identity of the target based on the topology structure of data. Obviously, the performance of inductive-learning-based classification method could depend on the number and quality of training samples more heavily than that of the transductive-learning-based method. As a result, we argue that transductive learning may be more suitable than inductive learning for FSL tasks. To effectively achieve few-shot SAR target identity reasoning, this article proposes a novel transductive-learning-based classification method named the transductive prototype reasoning. Specifically, we first calculate the initial class prototype for support samples belonging to the ith class with the following formula:

$$\mathbf{c}_i = \frac{1}{K} \sum_{m=1}^{K} \phi\left(\mathbf{x}_m^*\right) \tag{12}$$

where $\phi(\cdot)$ represents the feature embedding model.

Next, we adopt the following formula to predict the probability that the unlabeled query sample $\mathbf{x}$ belongs to the ith category target, i.e.,

$$w_{\mathbf{x},i} = \frac{\exp(-D(\phi(\mathbf{x}), \mathbf{c}_i))}{\sum_K \exp(-D(\phi(\mathbf{x}), \mathbf{c}_K))} \tag{13}$$

where $D(\cdot)$ represents the Euclidean distance metric function, which is defined as follows:

$$D(\phi(\mathbf{x}), \mathbf{c}_i) = \|\phi(\mathbf{x}) - \mathbf{c}_i\|_2^2. \tag{14}$$

Then, we construct a probabilistic confidence set for each category, denoted as $G_i, (i \in \{1, 2, \ldots, C\})$, which includes those unlabeled query samples whose probability weight is greater than the threshold ϵ, where C is the total number of SAR target categories.

According to the relevant knowledge of probability, we believe that if the probability of a sample belonging to the ith category is greater than 0.5, then the sample can be judged as this category of target; otherwise, the sample will not be labeled. The qualified query sample and corresponding probability weight will be put into the probabilistic confidence set, i.e.,

$$G_i = \left\{\left(\mathbf{x}, w_{\mathbf{x},i}\right) | w_{\mathbf{x},i} \geq 0.5\right\}, \quad i \in \{1, \ldots, C\}. \tag{15}$$

Finally, both the confidence set $G_i, (i \in \{1, 2, \ldots, C\})$ and support set S are used to update the initial SAR class prototype, aiming to generate a more representative SAR class prototype $\mathbf{c}_i{}^*$ according to the following expression:

$$\mathbf{c}_i{}^* = \frac{K\mathbf{c}_i + \sum_{n_i} w_{\mathbf{x},i} \cdot \phi(\mathbf{x})}{K + \sum_{n_i} w_{\mathbf{x},i}}, \quad \mathbf{x} \in G_i \tag{16}$$

where n_i is the number of samples in G_i.

Based on the class prototype calculated via (16), the probability of query sample belonging to the ith category is calculated by the Euclidean distance metric, i.e.,

$$p(i|\mathbf{x}) = \frac{\|\mathbf{x} - \mathbf{c}_i{}^*\|_2^2}{\sum_{k=1}^{C} \|\mathbf{x} - \mathbf{c}_k{}^*\|_2^2}. \tag{17}$$

D. Marginal Adaptive Hybrid Loss

To obtain a discriminative embedding space with intraclass compaction and interclass divergence, a new loss function, called MAHL consisting of classification loss and embedding loss, is presented, to train the proposed TPARN. Among them, cross-entropy loss is adopted as the classification loss, which is expressed as

$$L_c = -\frac{1}{CN} \sum_{i=1}^{C} \sum_{j=1}^{N} Y_{\mathbf{x},i} \log(p(y = i|\mathbf{x})) \tag{18}$$

where N represents the number of samples of the ith category SAR target. $Y_{\mathbf{x},i} = 1$ if the sample $\mathbf{x}$ belongs to class ith; otherwise, $Y_{\mathbf{x},i} = 0$, and $p(i|\mathbf{x})$ represents the predicted probability that sample $\mathbf{x}$ belongs to the ith category.

To learn a feature embedding space with intraclass compactness and interclass divergence, an embedding loss named marginal adaptive loss is presented in this article. Intuitively, the proposed marginal adaptive loss aims to force each sample to be close to positive prototype (i.e., the SAR category prototype of the same category) and meanwhile to be far from the nearest negative prototype (i.e., the SAR class prototype of different categories) by an adaptive threshold m. Mathematically, the marginal adaptive loss is defined as

$$L_m = \sum_{CN} \max\left(D\left(\phi(\mathbf{x}), \mathbf{c}_i{}^*\right) - \min_{j \neq i} D\left(\phi(\mathbf{x}), \mathbf{c}_j{}^*\right) + m, 0\right) \tag{19}$$

where $D(\phi(\mathbf{x}), \mathbf{c}_i{}^*) = \|\phi(\mathbf{x}) - \mathbf{c}_i{}^*\|^2$ represents the Euclidean distance between the sample $\mathbf{x}$ and the positive category prototype, and $D(\phi(\mathbf{x}), \mathbf{c}_j{}^*) = \|\phi(\mathbf{x}) - \mathbf{c}_j{}^*\|^2$ represents the Euclidean distance between the sample $\mathbf{x}$ and the negative category prototype. m is an adaptive threshold that represents the maximum distance between the positive and negative prototypes. Herein, m is calculated according to the following formula:

$$m = \max\left(\|\mathbf{c}_j{}^* - \mathbf{c}_i{}^*\|^2, j \neq i\right). \tag{20}$$

The total loss of the proposed TPARN is the weighted sum of classification loss and embedding loss, i.e.,

$$L = L_c + \lambda L_m \tag{21}$$

where λ is a weight that balances the importance of two losses.

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TABLE I
DETAILED INFORMATION OF EACH TARGET IN META-TRAINING STAGE AND META-TEST STAGE

Stage	Depression	Class	Number
Train stage	17°	BMP2	233
		BTR60	256
		BTR70	233
		T62	299
		T72	232
		D7	299
		ZIL131	299
Test stage	15°	2S1	274
		BRDM2	274
		ZSU23/4	274
	17°	2S1	299
		BRDM2	298
		ZSU23/4	299
	30°	2S1	288
		BRDM2	287
		ZSU23/4	288

TABLE II
FIVE DIFFERENT EXPERIMENTAL SCENARIOS

Experiment No.	Depression			
	Training stage		Test stage	
	Support	Query	Support	Query
1	17°	17°	15°	15°
2			17°	15°
3			17°	17°
4			17°	30°
5			30°	30°

TABLE III
SUPPORT SET AND TEST SET IN EXPERIMENTS

Setting	Stage			
	Train stage		Test stage	
	Support	Query	Support	Query
1-shot	3*1=3	3*10=30	3*1=3	3*10=30
5-shot	3*5=15	3*10=30	3*5=15	3*10=30

IV. EXPERIMENTS AND RESULTS

A. Dataset Description

In this section, we conduct a series of experiments on the moving and stationary target acquisition and recognition (MSTAR) benchmark dataset to assess the effectiveness of the proposed TPARN. The MSTAR dataset was publicly released by the U.S. Defense Advanced Research Projects Agency (DARPA). The SAR images in the MSTAR dataset were collected by the U.S. Air Force Research Laboratory using a 10-GHz X-band spotlight SAR sensor. The resolution of each SAR image is 0.3 × 0.3 m in both range and azimuth. The public MSTAR dataset includes ten categories of ground military targets: BTR60, BTR70, BMP2, T72, D7, T62, ZIL131, 2S1, BRDM2, and ZSU23/4. SAR images in the MSTAR dataset were acquired at various depression angles (e.g., 15°, 17°, and 30°) over the full aspect angles from 0° to 360°. Fig. 4 shows some SAR images and their corresponding optical images in the MSTAR dataset. Considering that the target in the original SAR image only occupies a few pixels of the image, all SAR images are cropped to 64 × 64 pixels RoI.

Because the proposed method adopts the idea of meta-learning, the MSTAR dataset needs to be divided into the meta-training dataset and meta-test dataset. Referring to earlier works [34], [35], [37], [38], [39], SAR images of seven targets, i.e., BTR60, BTR70, BMP2, T72, T63, ZIL131, and D7, acquired at 17° depression angle are selected as the meta-training dataset, and other three categories of military targets including 2S1, BRDM2, and ZSU23/4 collected at 15°, 17°, and 30° depression angle are used for the meta-test dataset, respectively. Detailed dataset information is listed in Table I.

B. Experimental Settings

To deliver convincing experimental results, we set up five different experimental scenarios in this article. Details are shown in Table II. In each scenario, two few-shot recognition tasks, i.e., three-way one-shot and three-way five-shot, are simulated. Meanwhile, referring to the existing works [36], [39], the detailed data configuration for different few-shot tasks in each scenario is listed in Table III. To obtain statistically significant experimental results, each experiment is conducted independently for 1000 runs.

This article adopts the Adam optimizer to train the proposed method. The learning rate is set to 0.001. The hyperparameter λ is set to 0.5 in the following experiments. The proposed method is implemented with the PyTorch framework, and all the experiments are carried out on a person computer with hardware configuration of Intel[1] Core[2] i7-9750 CPU at 2.60 GHz, 24.0-GB RAM NVIDIA GeForce RTX3090 GPU.

C. Recognition Results of TPARN

This section comprehensively evaluates the effectiveness of the proposed TPARN in five different scenarios. The experimental results are listed in Table IV, in which four evaluation protocols, i.e., maximum (Max), minimum (Min), average accuracy (Avg), and standard deviation, are provided. In addition, the confusion matrix of the average accuracy of the proposed TPARN under different experimental settings is presented in Table V, where TG1–TG3 represent 2S1, BRDM2, and ZSU23/4, respectively. To visually observe the robustness of the proposed TPARN, the statistical histogram of recognition accuracy for 1000 runs is plotted in Fig. 5.

As shown in Table IV, the average accuracy of the five groups of experiments can reach more than 80% under the one-shot setting. Surprisingly, the average accuracy of the five groups of experiments is above 85% under the five-shot setting. It is worth mentioning that the recognition accuracy

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Fig. 4. Some SAR images and their corresponding optical images. (a) BTR60, (b) BTR70, (c) BMP2, (d) T72, (e) D7, (f) T62, (g) ZIL131, (h) 2S1, (i) BRDM2, and (j) ZSU23/4.

TABLE IV
STATISTICAL RESULTS OF RECOGNITION ACCURACY OF TPARN

Experiment	Setting	Accuracy(%)			Standard deviation
		Max	Min	Avg	
1	1-shot	100.00	50.00	88.63	0.1075
	5-shot	100.00	77.00	94.20	0.0452
2	1-shot	100.00	30.00	80.67	0.0972
	5-shot	100.00	60.00	85.27	0.0637
3	1-shot	100.00	33.00	89.66	0.0943
	5-shot	100.00	73.00	94.07	0.0442
4	1-shot	100.00	37.00	85.43	0.0996
	5-shot	100.00	67.00	90.35	0.0557
5	1-shot	100.00	30.00	80.01	0.1056
	5-shot	100.00	57.00	88.03	0.0679

TABLE V
CONFUSION MATRIX OF AVERAGE RECOGNITION ACCURACY OF EACH GROUP OF EXPERIMENTS

No		Setting					
		1-shot			5-shot		
		TG1	TG2	TG3	TG1	TG2	TG3
1	TG1	**87.93**	5.89	6.18	**94.36**	2.90	2.74
	TG2	5.08	**89.23**	5.69	2.98	**94.12**	2.90
	TG3	5.93	5.39	**88.68**	2.78	3.14	**94.08**
2	TG1	**82.02**	8.98	9.00	**85.28**	7.41	7.31
	TG2	9.26	**80.96**	9.78	7.78	**85.01**	7.21
	TG3	8.94	8.99	**82.07**	7.56	6.92	**85.52**
3	TG1	**89.09**	5.70	5.21	**94.1**	2.81	3.09
	TG2	5.03	**89.95**	5.02	3.19	**93.96**	2.85
	TG3	5.27	4.82	**89.91**	2.76	3.08	**94.16**
4	TG1	**85.87**	6.83	7.30	**90.13**	4.89	4.98
	TG2	7.38	**85.55**	7.07	3.82	**91.01**	5.17
	TG3	7.69	7.45	**84.86**	4.21	5.87	**89.92**
5	TG1	**79.73**	10.07	10.2	**87.68**	5.14	7.18
	TG2	11.04	**79.84**	9.12	5.11	**88.07**	6.82
	TG3	9.68	9.85	**80.47**	5.56	6.11	**88.33**

of the proposed method is up to 90% under the five-shot setting of experiments 1, 3, and 4. However, there is a seemingly anomalous phenomenon where the performance of the proposed method degrades under the two experimental settings of experiment 5. As a matter of fact, this is understandable because the large depression difference between the meta-training dataset and meta-test dataset leads to the obvious difference in SAR image features, which gives rise to performance degradation. Still and all, the average accuracy under the one-shot and five-shot settings of experiment 5 can still reach 80.01% and 88.03%, respectively. Moreover, it can be seen from Table IV that the average accuracy of the proposed method under the five-shot setting is higher than that of the one-shot setting, and the standard deviation under the five-shot setting is lower than that of the one-shot setting. This is consistent with the statistical histogram distribution of recognition rate shown in Fig. 5. From the confusion matrices in Table V, one can see that the recognition performance of the proposed method on three categories of targets is balanced under different experimental settings. Therefore, it follows that the proposed method has huge potential in solving the few-shot SAR target recognition task.

D. Comparison to Other Few-Shot SAR ATR Methods

To illustrate the superiority of the proposed method in the few-shot SAR recognition task, this section conducts a series of comparative experiments. The comparison methods are divided into two categories, i.e., inductive-learning-based methods and transductive-learning-based method. Among them, the inductive-learning-based methods include ProtoNet [23], RelationNet [24], metalearning SAR (MSAR) [36], cross attention network (CAN) [42], MTRLF [38], 2SCNet [39], and Bi-similarity with capsule embedding network (BSCapNet) [34]. Two transductive-learning-based methods, i.e., TPN [26] and transductive CAN (TCAN), are also used as competitors. The experimental results are listed in Table VI, where "Trans." in the third column indicates whether the method is based on transductive learning. "Yes" represents that this method is based on transductive learning, and "No" indicates that this method is based on inductive learning. As can be observed from Table VI, the proposed TPARN always outperforms all the inductive-learning-based competitors under different few-shot SAR target recognition tasks. Especially in experiment 5, the recognition performance of the proposed TPARN is always superior to all the inductive-learning-based competitors under large depression variations. In view of this phenomenon, it can be concluded that the transductive-learning-based

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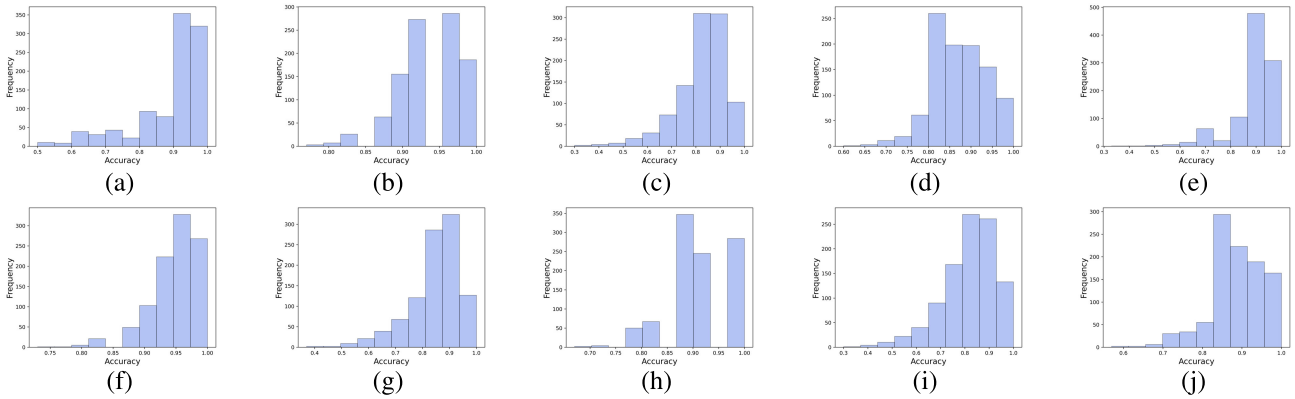


Fig. 5. Statistical histogram of recognition rate for 1000 runs of independent experiments on the MSTAR dataset. (a) Experiment 1 one-shot. (b) Experiment 1 five-shot. (c) Experiment 2 one-shot. (d) Experiment 2 five-shot. (e) Experiment 3 one-shot. (f) Experiment 3 five-shot. (g) Experiment 4 one-shot. (h) Experiment 4 five-shot. (i) Experiment 5 one-shot. (j) Experiment 5 five-shot.

TABLE VI
PERFORMANCE COMPARISONS OF THE PROPOSED TPARN AND SOME ADVANCED FEW-SHOT METHODS

Setting	Method	Trans.	Experiment				
			1	2	3	4	5
1-shot	ProtoNet	No	71.51±0.65	66.81±0.66	69.90±0.62	64.05±0.65	66.76±0.79
	RelationNet	No	74.22±0.67	67.50±1.22	69.95±0.62	64.41±0.70	62.78±0.84
	CAN	No	77.36±0.65	70.31±0.76	75.79±0.80	72.82±0.86	78.30±0.85
	MSAR	No	67.71±0.85	66.59±0.85	68.43±0.88	63.71±1.04	64.51±1.12
	2SCNet	No	78.02±0.68	70.02±0.63	72.14±0.66	68.21±0.62	68.26±0.71
	MTRLF	No	86.11±0.70	74.32±0.79	81.59±0.89	78.20±0.73	70.17±0.76
	BSCapNet	No	72.01±0.80	70.84±0.71	74.37±0.65	73.98±0.47	67.22±0.68
	TPN	Yes	75.51±0.83	70.79±0.69	77.48±0.69	73.70±0.73	68.26±0.70
	TCAN	Yes	79.44±0.73	72.73±0.82	77.96±0.91	75.47±0.87	78.71±0.94
	TPARN	Yes	**88.63±0.67**	**80.67±0.60**	**89.66±0.58**	**85.43±0.62**	**80.01±0.67**
5-shot	ProtoNet	No	80.28±0.42	81.29±0.41	82.22±0.42	78.24±0.43	78.59±0.49
	RelationNet	No	79.54±0.40	80.46±0.42	81.21±0.39	74.36±0.43	74.60±0.57
	CAN	No	85.53±0.40	77.42±0.47	84.70±0.51	82.24±0.47	86.57±0.52
	MSAR	No	82.62±0.58	82.21±0.63	83.83±0.61	77.89±0.67	77.55±0.69
	2SCNet	No	84.21±0.40	82.40±0.41	84.91±0.39	81.08±0.41	81.13±0.45
	MTRLF	No	93.78±0.50	82.72±0.56	91.23±0.43	87.22±0.67	78.72±0.52
	BSCapNet	No	82.65±0.45	78.71±0.47	86.60±0.48	83.20±0.44	80.51±0.49
	TPN	Yes	82.87±0.56	76.98±0.55	85.04±0.52	80.65±0.55	79.31±0.48
	TCAN	Yes	85.89±0.41	78.52±0.52	85.75±0.52	82.46±0.52	86.92±0.43
	TPARN	Yes	**94.20±0.28**	**85.27±0.39**	**94.07±0.27**	**90.35±0.35**	**88.03±0.42**

method is more suitable than the inductive-learning-based method for solving the few-shot SAR target recognition task.

Compared with the transductive-learning-based few-shot methods, the recognition rate of the proposed TPARN in the first four experiments is about 7% higher than that of all the competitors. Although the recognition performance improvement of the proposed TPARN in the fifth experiment is not as obvious as that in the first four experiments, the proposed method still outperforms all the competitors under two different settings. It is worth noting that unlike other transductive-learning-based methods, the proposed TPARN only uses those samples with high confidence rather than all the query samples, which can reduce the risk of wrong reasoning due to sample mislabeling to some extent. Overall, the proposed TPARN shows outstanding performance in solving the problem of few-shot SAR target recognition.

E. Ablation Studies

In this section, a series of experiments are performed to verify the effectiveness of each key component of the proposed TPARN. All the ablation experiments are conducted under five different SAR scenarios.

First, we deploy experiment to investigate the efficacy of region-awareness-based feature extractor in the proposed TPARN. For the sake of simplicity, we replace the region-awareness-based feature extractor with the feature extraction network of ProtoNet, which is termed as model 1, and the proposed TPARN is termed as model 2. The experimental results

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TABLE VII
ABLATION EXPERIMENTS FOR EFFECTIVENESS ASSESSMENT OF REGION-AWARENESS-BASED FEATURE EXTRACTOR

Setting	Model	Experiment				
		1	2	3	4	5
1-shot	1	82.41	74.87	82.36	78.43	70.05
	2	**88.63**	**80.67**	**89.66**	**85.43**	**80.01**
5-shot	1	88.14	82.18	91.08	87.12	80.06
	2	**94.20**	**85.27**	**94.07**	**90.35**	**88.03**

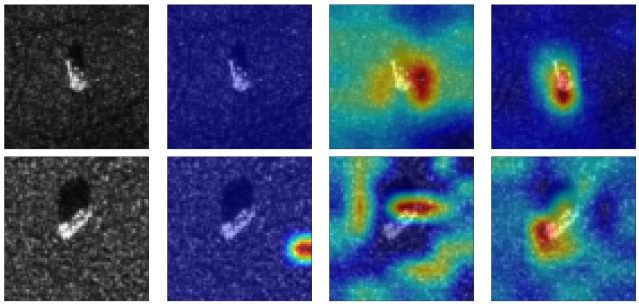


Fig. 6. Feature visualization results of three models. (a) Original SAR image. (b) Features' map from CNN. (c) Features' map from SENet. (d) Features' map from the proposed TPARN.

TABLE VIII
ABLATION EXPERIMENTS FOR EFFECTIVENESS ASSESSMENT OF CFSA MODULE

Setting	w/o	Experiment				
		1	2	3	4	5
1-shot	o	82.29	75.75	83.71	80.79	73.76
	w	**88.63**	**80.67**	**89.66**	**85.43**	**80.01**
5-shot	o	89.43	83.78	91.74	88.15	83.90
	w	**94.20**	**85.27**	**94.07**	**90.35**	**88.03**

TABLE IX
ABLATION EXPERIMENTS FOR EFFECTIVENESS ASSESSMENT OF TRANSDUCTIVE PROTOTYPE REASONING

Setting	w/o	Experiment				
		1	2	3	4	5
1-shot	o	84.04	77.91	85.78	82.47	75.38
	w	**88.63**	**80.67**	**89.66**	**85.43**	**80.01**
5-shot	o	93.18	83.47	93.02	89.25	86.69
	w	**94.20**	**85.27**	**94.07**	**90.35**	**88.03**

are given in Table VII. It can be observed from Table VII that the recognition rate of model 2 is always higher than that of model 1. Therefore, it follows that the region-awareness-based feature extractor is capable of extracting more transferable knowledge, which is very beneficial to improve the few-shot SAR target recognition performance.

Besides, to intuitively investigate the proposed region awareness feature extraction model can better locate the RoI in SAR images. We conduct several qualitative analysis experiments in this section. A visualization method, i.e., GRAD-CAM [55], is leveraged to display the extracted feature map. To be exact, GRAD-CAM uses gradients to determine the importance of spatial locations, so as to reflect those regions that the model considers important for the predicted tasks. Herein, CNN [23] and SENet [43] are used as competitors. Fig. 6 shows the visualization results of different methods, where the first row is the result of one image of T62 and the second row is the result of one image of BTR60. These experimental results in Fig. 6 show that the proposed region awareness feature extractor focuses more on the target region than the two comparison methods. In summary, the proposed region awareness feature extractor can better extract the RoI features in SAR images, thereby improving the performance of the recognition model.

Next, we carry out several experiments to assess the effectiveness of the CFSA module in promoting the few-shot SAR target recognition performance. Table VIII lists all the experimental results, in which "w" represents that the proposed method includes the CFSA module, and "o" represents that the proposed method removes the CFSA module.

Apparently, the recognition rate of the method without the CFSA module is inferior to the proposed TPARN under five different SAR scenarios. These experiments demonstrate that the CFSA module is helpful to improve the recognition performance of the model. Herein, the experimental results give us an inspiration that spatial attention derived from CFF can heighten the discriminative ability of the extracted features, so as to improve the performance of few-shot SAR target recognition.

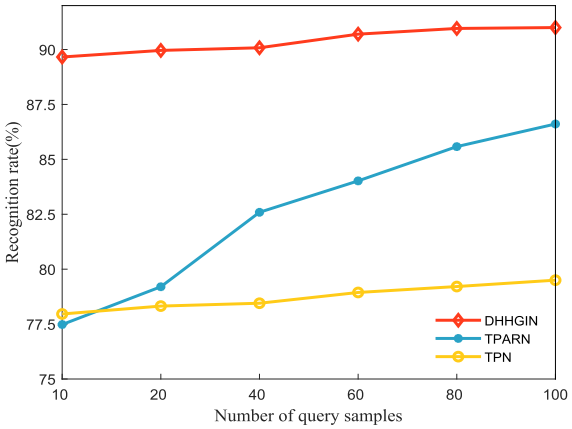
Then, we pursue several experiments to evaluate the effectiveness of transductive prototype reasoning for few-shot SAR target recognition. Table IX presents the corresponding experimental results, in which "w" means that the proposed method is based on transductive learning, and "o" means that the proposed method is based on inductive learning. As can be observed from Table IX, compared with the inductive-learning-based method, the recognition rate of the transductive prototype reasoning method is improved by about 3% and 1% under the one-shot and five-shot settings, respectively. From these convincing experimental results, we can state that the proposed transductive prototype reasoning method is better than some state-of-the-arts to solve the problem of few-shot SAR target recognition.

We finally investigate the effectiveness of the proposed MAHL function. Table X gives the experimental results, in which CE represents the cross-entropy loss function is used for model training, and MAHL indicates that MAHL is selected for model training. From Table X, one can see that the recognition performance of the method using MAHL is always better than that of the method using CE loss. One main reason is that with the help of MAHL, the proposed TPARN can learn a discriminative feature embedding space with intraclass compactness and interclass divergence, which is very beneficial to improve the recognition performance.

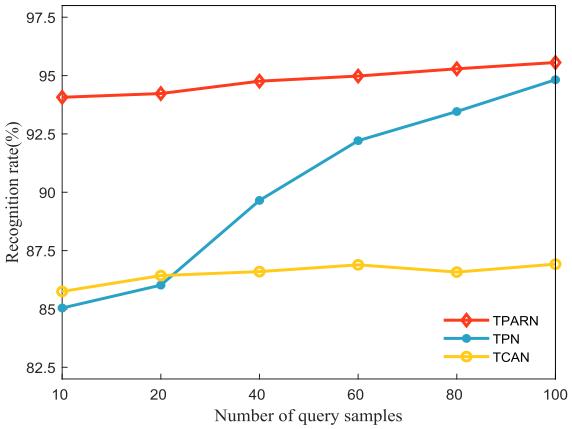
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TABLE X
ABLATION EXPERIMENTS FOR EFFECTIVENESS ASSESSMENT OF MAHL

Setting	Loss Function	Experiment 1	2	3	4	5
1-shot	CE	87.14	80.03	87.92	84.69	73.27
	MAHL	**88.63**	**80.67**	**89.66**	**85.43**	**80.01**
5-shot	CE	93.40	83.80	93.12	89.40	83.00
	MAHL	**94.20**	**85.27**	**94.07**	**90.35**	**88.03**



(a)



(b)

Fig. 7. Graph of the change in recognition rate with the number of query samples. (a) Experiment 3 one-shot. (b) Experiment 3 five-shot.

F. Transductive Reasoning Experiment

Considering that the proposed method is based on the idea of transductive learning, the number of query samples can inevitably affect the recognition performance of the model. As a consequence, this section explores the impact of the number of query samples on the model recognition performance. To illustrate the superiority of the proposed TPARN, two state-of-the-art transductive learning methods, i.e., TPN and TCAN, are used as competitors. Fig. 7 depicts the graphs of the recognition rate versus the number of query samples for the three methods under the one-shot and five-shot settings of experiment 3, respectively.

From Fig. 7, it can be seen intuitively that the recognition rate of the proposed TPARN and the two competitors increases to varying degrees with the increase in the number of query samples. It is particularly noteworthy that when the query sample changes from 100 to 10, the performance of the proposed TPARN changes gently, while the two competitors vary dramatically. In addition, one can see from Fig. 7 that the recognition rate of the proposed method is always superior to the two competitors. As a result, we can state that the proposed TPARN is effective and robust to solve the problem of few-shot SAR target recognition.

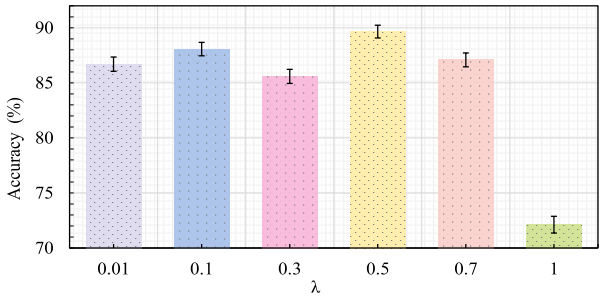


Fig. 8. Recognition rate of the proposed method with different values of λ.

G. Hyperparameter λ Analysis

λ is a key parameter that balances the importance of classification loss and embedding loss in the total loss function of the proposed method. To investigate the influence of the value of λ on the recognition performance of the proposed method, this section adopts the enumeration method to pursue the related experiments. All the experiments are conducted in experimental scenario 3, as listed in Table II. The value of λ is $\{0.01, 0.1, 0.3, 0.5, 0.7, 1\}$ in this experiment. Fig. 8 displays the recognition rate of the proposed method with different λ values. These experimental results in Fig. 8 show that the recognition rate of the proposed method increases first and then decreases with the change in the value of λ, which indicates that an appropriate hyperparameter λ is important for the model. According to the experimental results in Fig. 8, it is reasonable to set λ to 0.5 in this article.

V. CONCLUSION

This article proposes a novel method named TPARN to solve few-shot SAR target recognition. The innovation of this article includes four aspects. First, a region-awareness-based feature extraction network is designed to extract more transferable knowledge for few-shot SAR target recognition. Second, a CFSA module is proposed to heighten the discriminative ability of the extracted features, aiming to improve the few-shot SAR target recognition performance. Third, a transductive prototype reasoning method is presented to robustly achieve SAR target identity reasoning. Fourth, a new loss called MAHL is proposed to learn a discriminative feature embedding network with intraclass compactness and interclass divergence. Experiments on the MSTAR benchmark dataset demonstrate that the proposed method is superior to several state-of-the-arts for few-shot SAR target recognition.

In the following research, we will further explore the few-shot SAR recognition algorithm under open-set conditions.

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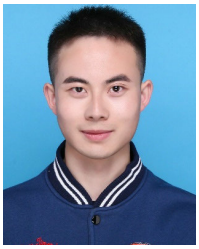
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Haohao Ren (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2021.

From 2020 to 2021, he was a Visiting Ph.D. Student with the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento, Trento, Italy. Currently, he is an Assistant Fellow with the School of Information and Communication Engineering, UESTC. His research interests include radar signal processing, synthetic aperture radar (SAR), target recognition, pattern recognition, deep learning, and Bayesian learning.



Sen Liu (Student Member, IEEE) received the B.S. degree from the School of Electronics and Information Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2020, where he is currently pursuing the master's degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar, automatic target recognition, and deep learning.



Xuelian Yu (Member, IEEE) received the B.S. and M.S. degrees in signal and information processing from the Xi'an University of Science and Technology, Xi'an, China, in 2001 and 2004, respectively, and the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2008.

From 2008 to 2010, she held a post-doctoral position at the Department of Electronic Engineering, UESTC. From 2014 to 2015, she was a Visiting Scholar with the Department of Electronic Engineering, The Ohio State University, Columbus, OH, USA. Since 2010, she has been an Associate Professor with the School of Information and Communication Engineering, UESTC. Her research interests include radar signal processing, synthetic aperture radar, target recognition, target tracking, and manifold learning.



Lin Zou (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in signal and information processing from the School of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1999, 2002, and 2013, respectively.

From 2014 to 2015, he was a Visiting Scholar with the Department of Electronic Engineering, Linköping University, Linköping, Sweden. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar signal processing, radar systems, and radar echo simulator.



Yun Zhou received the Ph.D. degree in signal and information processing from the School of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2016.

He is currently a Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar signal processing, target recognition, and target tracking.



Xuegang Wang received the B.S., M.S., and Ph.D. degrees in signal and information processing from the School of Electronic Engineering, Xidian University, Xi'an, China, in 1984, 1987, and 1992, respectively.

From 1992 to 1994, he held a post-doctoral position at the Department of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, where he has been a Professor with the Department of Electronic Engineering since 2000. He is currently the Director of the Laboratory of Radar System and Digital Technology. His research interests include radar signal processing, target recognition and detection, radar echo simulator, and statistical signal processing.



Hao Tang received the master's degree from the School of Electronic and Computer Engineering, Peking University, Beijing, China, in 2016, and the Ph.D. degree from the Multimedia and Human Understanding Group, University of Trento, Trento, Italy, in 2021.

He was a Visiting Scholar with the Department of Engineering Science, University of Oxford, Oxford, U.K. He is currently a Post-Doctoral Researcher with the Computer Vision Laboratory, ETH Zürich, Zürich, Switzerland. His research interests are deep learning, machine learning, and their applications to computer vision.

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